Tankala Sai Prasanth

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Experience

SOFTWARE ENGINEER | AI & POWER PLATFORM DEVELOPER | LTM | AUGUST 2025 – PRESENT

- Architected and deployed enterprise autonomous AI agents using Microsoft Copilot Studio and generative AI models, engineering multi-turn orchestration, agentic reasoning, and retrieval-augmented generation (RAG) across corporate data repositories.
- Designed custom Copilot agent actions and extensible plugins using Azure AI services, automating multi-step customer and internal workflows through conversational AI interfaces.
- Spearheaded end-to-end Power Platform solutions utilizing Power Apps (Canvas & Model-Driven), Power Automate, and Dataverse, automating complex enterprise workflows and slashing manual processing time by 50%.
- Engineered multi-stage automated cloud flows with Power Automate custom connectors and REST APIs, eliminating ~20 manual operational hours per week.
- Developed scalable backend microservices and secure data persistence layers using Java 17+, Spring Boot, and optimized SQL schemas to power API integrations for enterprise AI agents and business applications.

MACHINE LEARNING INTERN | INTRAIN TECH | OCTOBER 2023 – NOVEMBER 2023

- Built Python-based machine learning models for credit risk scoring and loan validation, achieving a **15% reduction** in prediction error rates.
- Improved model evaluation metrics by **20%** through iterative hyperparameter tuning, advanced feature engineering, and ensemble methods.

Skills

- **AI & Agentic Systems:** Agentic AI, Autonomous AI Agents, Multi-Agent Orchestration, Microsoft Copilot Studio, Azure AI Studio, Claude API, Google Gemini API, Prompt Engineering, RAG Pipelines, Vector Search & Embeddings, Reinforcement Learning (PPO, SAC), Responsible AI.
- **Data Engineering & Analytics:** Data Modeling, ETL/Data Ingestion Pipelines, Data Preprocessing, Data Analytics & Visualization, Exploratory Data Analysis (EDA), Feature Engineering, Pandas, NumPy, Power BI.
- **Databases & Cloud Storage:** Dataverse, SQL (PostgreSQL, MySQL / Schema Tuning & Query Optimization), MongoDB (NoSQL), Vector Databases (ChromaDB / FAISS concepts), AWS Cloud Infrastructure, Azure Cloud.
- **Programming & Systems:** Python, Java (17+), FastAPI, Spring Boot, Spring Security, REST APIs, WebSockets, Git, GitHub, ReactJS, Data Structures & Algorithms, Telemetry & Performance Tuning, ALM.

Projects

AutoFoos: Hierarchical Multi-Agent System (HMAS)

- Architected a real-time web-based Hierarchical Multi-Agent System (HMAS) orchestrating 10 autonomous physics agents in dynamic competitive simulation.
- Implemented supervisory LLM coaches powered by Google Gemini to analyze opponent shot spatial clusters and issue real-time strategic directives.
- Developed reactive rod agents using raycasting for real-time ball interception, dynamic passing corridors, and defensive anchoring.
- Built a headless 60 Hz physics engine in FastAPI streaming real-time simulation state over bi-directional WebSockets to an HTML5 Canvas interface. (Live Demo | GitHub Repo)

Digital Twin Technology for Diabetes Risk Prediction

- Built an AI-driven digital twin model integrating policy-gradient reinforcement learning (PPO, SAC) and XGBoost, achieving 94.5% prediction accuracy for Type 2 diabetes risk trajectories and adaptive health recommendations.

NeuroVision: AI-Driven Brain Tumor Detection from MRI Images

- Engineered a deep learning diagnostic system utilizing CNN, VGG16, and U-Net architectures on MRI scans, achieving 97.14% classification accuracy and sub-2-second inference latency deployed via Flask and React.

Education

Bachelor of technology, VIT-AP University **Amaravathi, AP** **12/2021 – 04/2025**

- B. Tech in computer science and business systems with a CGPA of 8.95/10.

Publications

"AI-Enhanced Smartwatches: Bridging Technology, Behavior, and Continuous Usage" *IEEE*, September 2025 | IEEE Xplore: 11158655

- *Co-Author* — Formulated research methodology, led empirical model experimentation and behavioral telemetry analysis, and contributed to the peer-reviewed conference manuscript.

Certifications & Publication

- Microsoft Certified: Azure AI Apps and Agents Developer Associate (Issued Oct 2026 · Expires Oct 2027)
- Anthropic: Claude Certified Developer (Issued Aug 2026 · Expires Aug 2027)
- Microsoft Certified: Agentic AI Business Solutions Architect | Microsoft (August 2026)
- Microsoft Certified: AI Agent Builder Associate | Microsoft (August 2026)
- AWS Certified: AI Practitioner | AWS (June 2026) & AWS Cloud Practitioner | AWS (August 2026)